Hearth Hold - Technical Sheet

Overview

Genre: colony sim with settlers, needs and raids (original take). One self-contained HTML file, about 31 KB, no external assets, libraries or network calls. Built for phones first, portrait layout.

Tech used

HTML5 Canvas 2D for all graphics, drawn in code: layered trees that sway, shaded rocks, growing crops, huts with glowing windows at night, flickering Hearth fire with embers, towers with pulsing crystals.

Day/night cycle (60 s days) with a night tint, additive light pools around the Hearth, huts and towers, and fireflies.

Particles, tower bolts, hit sparks, screen shake and floating resource numbers.

Web Audio API: all sound and music generated live. A calm triangle-wave folk loop by day, darker filtered at night, with a drum pulse during raids. Sound effects for chopping, building, horns, bolts and deaths.

Pointer Events for touch and mouse (tap and drag), large 56 px buttons, safe-area insets, devicePixelRatio scaling. localStorage auto-save and best-day record.

Game systems

12 x 18 tile valley, seeded random terrain. A job AI per settler picks the most urgent task: fight nearby raiders, eat, sleep, harvest, build, then chop or mine, with no two settlers on the same tile.

Needs: hunger, rest, mood and health, with starvation damage, mood breaks and grief when someone dies.

Economy: food, wood, stone; farms grow only in daylight; trees slowly regrow.

Raids: size 1 + day/2.7, health 30 + 4.4 per day, damage 5 + 0.6 per day. Raiders chase settlers within 2 tiles, otherwise march on the Hearth and break any building in the way.

Population cap 3 + 2 per hut; new settlers arrive at dawn if food allows.

Audit 1: Balance (simulated play)

A bot player (farms per settler, huts, towers, walls, marking, rallying during raids) played 20 full colonies per round, plus an idle player that never acts.

First version: the bot reached day 30 in 19 of 20 colonies, too easy. Raids were scaled up (size and health); then 0 of 20 won, too hard.

Final tuning: average colony lasts about 30 days, about half reach day 30, none fall before day 8. An idle player lasts exactly 10 days, so actions clearly matter. Difficulty climbs steadily as raids grow every night.

Audit 2: Bugs

A headless test plays the full UI: every tool, invalid taps, rally, help, 8 in-game days, save and reload, then plays to a game over. Final result: 0 errors.

Fixed: badly hurt settlers kept fighting to the death; they now back off below 25 health.

Fixed: during raids every settler re-chose a job every frame, making idle settlers jitter; it now re-checks twice a second.

Fixed: dragging the Farm or Hut tool could place many buildings by accident; drag only works for marking and walls.

Audit 3: Player experience

Added "raid tonight!" in the top bar the day before a raid, a raid horn and a dusk banner.

Added tips on day 2 (build a hut) and day 3 (build a tower), and a red food counter when food runs low.

Settler list shows each person's state at a glance (hungry, tired, hurt, miserable, happy).

Clear messages for invalid taps (wrong tile, not enough materials).

Testing note

Tested with simulated play and a headless browser harness, not on a physical phone.